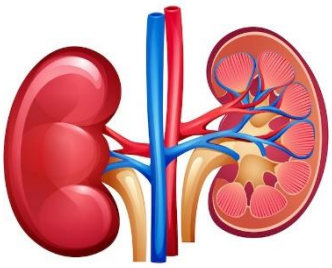

Kidney Disease Image Binary Classification

Data Science and Engineering

Data Science Project, 4th year, Data Science and Engineering

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1. Abstract

Kidney disease represents a significant global health issue, and early detection through non-invasive imaging can greatly improve patient outcomes. In this project, we developed a binary image classification system capable of distinguishing between healthy and diseased kidneys based on ultrasound images. The dataset used included 1,985 renal ultrasound images—1,535 labeled as diseased and 450 as healthy—making the dataset both small and highly imbalanced.

We explored three state-of-the-art convolutional neural network (CNN) architectures: MobileNetV2, ResNet50, and EfficientNetB0. Each model underwent a comprehensive transfer learning pipeline, including progressive layer unfreezing, threshold optimization using ROC curve analysis, and performance tuning based on F1-score and balanced accuracy. The best-performing model, ResNet50, achieved an AUC of 0.84 and demonstrated strong generalization across both classes.

In parallel with model development, we designed and implemented a full-stack web application that allows users to upload new renal ultrasound images along with their corresponding masks and obtain real-time predictions from the trained models. This app integrates all three final models and offers a user-friendly interface, model selection, and PDF report generation for medical professionals.

The results show that deep learning techniques, combined with transfer learning and thoughtful model calibration, can offer robust tools for medical image classification even under limited data conditions. The deployed system demonstrates the feasibility of using automated AI-based screening tools for assisting kidney disease diagnosis, and serves as a foundation for future enhancements involving larger datasets, ensemble learning, and model explainability.

2. Introduction

Kidney diseases, including chronic kidney disease (CKD), affect over 850 million people worldwide and often remain undetected until reaching advanced stages. Early diagnosis is crucial to prevent disease progression and reduce the risk of kidney failure, cardiovascular complications, and mortality. Ultrasound imaging is one of the most accessible and non-invasive diagnostic tools available, making it a first-line examination method in nephrology. However, the manual interpretation of renal ultrasound images requires expert radiological knowledge and is prone to inter-observer variability, especially in early or subtle disease stages.

In recent years, deep learning has emerged as a powerful approach for automatic image interpretation. Convolutional Neural Networks (CNNs), in particular, have shown remarkable performance in medical image classification tasks. Nevertheless, training deep learning models in the healthcare domain often comes with specific challenges: limited and imbalanced datasets, high intra-class variability, and the need for model interpretability.

This project addresses the problem of automated classification of renal ultrasound images into two categories: healthy and diseased. The objective is to build a robust, high-performing binary classifier that can support medical professionals by providing reliable second opinions. To achieve this, we explore and compare three state-of-the-art CNN architectures—MobileNetV2, ResNet50, and EfficientNetB0—applying transfer learning techniques and advanced model optimization strategies.

Beyond model development, a core goal of this project was to create a complete web application that allows users to upload renal ultrasound images with their corresponding segmentation masks and receive a classification output in real time. This web application integrates the three final models and offers model selection, result visualization, and a downloadable diagnostic report in PDF format.

By combining high-level machine learning models with an interactive deployment interface, this project provides a comprehensive, scalable solution for supporting kidney disease diagnosis. The final system aims to bridge the gap between academic research and clinical utility, demonstrating the potential of AI-assisted tools in real-world healthcare settings.

3. Problem Statement and Objectives

3.1 Problem Definition

Kidney disease is a growing global health concern that often develops silently, with symptoms appearing only at advanced stages. Early detection through renal ultrasound imaging can significantly reduce morbidity and mortality rates by allowing timely intervention. However, several challenges hinder effective diagnosis:

- Interpretation Difficulty: Ultrasound images are highly dependent on operator skill, and interpreting them requires specialized training. The quality of diagnosis may vary between radiologists and across institutions.

- Data Constraints: Medical imaging datasets are often limited in size due to privacy concerns and acquisition difficulties. In our case, the dataset consisted of 1985 renal ultrasound images, with a significant class imbalance: 1535 diseased and 450 healthy cases. This imbalance can lead to biased predictions and reduced performance on the underrepresented class.

- Lack of Automation Tools: While machine learning has shown promise in medical diagnostics, few end-to-end solutions exist that are accessible to clinicians for real-time classification of renal images, especially in resource-constrained settings.

- Need for Interpretability: In medical applications, it is essential that automated tools are not only accurate but also interpretable and transparent in their decision-making process.

Given these challenges, there is a clear need for a system that can reliably classify renal ultrasound images while being accessible, interpretable, and easy to integrate into clinical workflows.

3.2 Objectives

To address the above challenges, the project is structured around the following key objectives:

Primary Objectives

- Develop a binary classification system capable of distinguishing between healthy and diseased kidneys using renal ultrasound images.

- Evaluate and compare three pre-trained deep learning architectures: MobileNetV2, ResNet50, and EfficientNetB0.

- Fine-tune models using transfer learning, threshold optimization, and advanced training strategies to improve performance on a small, imbalanced dataset.

Secondary Objectives

- Design and implement a web application that allows users to:
    - Upload new ultrasound images and their corresponding segmentation masks.
    - Select which model to use for classification.
    - Visualize classification probabilities and final diagnostic output.
    - Download a detailed PDF report with predictions and confidence levels.
- Conduct a thorough model performance analysis, including:
    - ROC curves and AUC comparison.
    - Per-class precision, recall, and F1-score.
    - Confusion matrices.
    - Comparative visualizations and threshold tuning effects.

Explore model explainability as a future enhancement, laying the groundwork for integrating tools like Grad-CAM to support clinical trust.

4. Methodology

This section describes the pipeline followed to develop, train, and deploy the classification system for kidney ultrasound images. The methodology can be divided into three main stages: data preprocessing, model development and training, and web application implementation.

4.1 Data Preprocessing

4.1.1 Dataset description

The dataset consisted of 1985 renal ultrasound images and their corresponding segmentation masks, sourced from real clinical cases. The class distribution was heavily imbalanced:

- Diseased (Kidney with pathology): 1535 images

- Healthy (No pathology): 450 images

This imbalance introduced biases in early models, leading them to favour the majority class (diseased), which necessitated advanced strategies during training and evaluation.

4.1.2 Preprocessing Pipeline

Each image underwent a series of preprocessing steps to standardize input and enhance model performance. These steps were applied identically across all three models (MobileNetV2, ResNet50, EfficientNetB0):

- Mask Overlay: Only the region of interest (the kidney) was preserved by applying the segmentation mask to the ultrasound image. This removed background noise and focused the model's attention.
- Resizing: All images were resized to match the input size required by each architecture:
    - MobileNetV2: 224x224
    - ResNet50: 224x224
    - EfficientNetB0: 224x224
- Colour Conversion: Since models were pre-trained on RGB images, grayscale images were duplicated across three channels.
- Normalization: Pixel values were scaled to the [0, 1] range.

- Shuffling and Stratification: The dataset was split into training, validation, and test sets using a 70-15-15 stratified ratio to preserve class balance in each subset.
- Image Augmentation: To improve generalization and mitigate overfitting, we applied transformations such as:
    - Horizontal and vertical flipping
    - Random brightness/contrast adjustment
    - Minor rotations and zooms

4.2 Model Development and Training

4.2.1 Transfer Learning Strategy

Three pre-trained convolutional neural network (CNN) models were selected for evaluation:

1. MobileNetV2
2. ResNet50
3. EfficientNetB0

These architectures were chosen for their balance between performance and computational efficiency, especially important given our relatively small dataset.

All models were initialized with ImageNet weights, and transfer learning was applied using the following strategy:

- Frozen Base Layers: Initially, all convolutional layers were frozen to train only the custom classification head.
- Progressive Unfreezing: After initial training, we gradually unfroze top layers of the base model, fine-tuning them with a reduced learning rate.
- Early Stopping and Checkpointing: Training was monitored with early stopping (patience of 10 epochs) based on validation loss, and best weights were saved.

4.2.2 Custom Classification Head

Each model shared a similar classification head structure:

- Global Average Pooling
- Dense Layer (128 or 256 units) with ReLU activation
- Dropout (rate 0.2–0.3)
- Final Dense Layer (1 neuron with sigmoid activation)

4.2.3 Loss Function and Optimization

- Loss Function: Binary Cross-Entropy
- Optimizer: Adam (learning rate: 1e-4 to 1e-5 during fine-tuning)

- Batch Size: 32
- Epochs: 100 (early stopping typically halted training earlier)

4.2.4 Threshold Optimization

Due to class imbalance and varying score distributions, the default threshold of 0.5 for classification was not ideal. Instead, we performed threshold optimization for each model based on F1-score and balanced accuracy, selecting a value that maximized performance on the validation set.

4.3 Web Application implementation

In addition to building the classification models, a fully functional web application was developed to demonstrate their practical utility.

4.3.1 Technologies Used

The system was built with a modular stack:

- Frontend: Developed in Next.js, a React-based framework, responsible for:
    - Uploading ultrasound images and segmentation masks
    - Displaying classification results and confidence levels
    - Allowing model selection from MobileNetV2, ResNet50, or EfficientNetB0
- Backend: Built in Python (Flask-like logic), responsible for:
    - Handling image preprocessing and prediction
    - Loading the appropriate .keras model file from the models/ directory
    - Returning prediction probabilities and class output
- Models Served: The three best fine-tuned models were saved in .keras format and loaded on demand.

4.3.2 Application Features

- Upload Interface: Users can submit both ultrasound and mask images.
- Model Selector: Choose between the three trained models.
- Visualization Panel: Displays the masked input, prediction probability, and final class label.
- Export Function: Download a detailed PDF report with:
    - Model used
    - Classification result
    - Confidence score
    - Uploaded image preview

4.3.3 Usage Flow

1. User uploads ultrasound + mask image.
2. Selects preferred model.
3. Receives prediction with confidence and downloadable report.

The application enhances accessibility for non-technical users and demonstrates how machine learning can be embedded in clinical tools.

5. Experimental Results and Evaluation

This section presents the evaluation of the three models trained: **MobileNetV2**, **ResNet50**, and **EfficientNetB0**, in terms of performance metrics, confusion matrices, and visual comparisons. Given the class imbalance in the dataset (77% diseased, 23% healthy), special attention was paid to metrics that account for both sensitivity and specificity.

5.1 Evaluation Metrics

The following metrics were computed for each model on the test set:

- Accuracy: Overall proportion of correctly classified images.
- Precision: Correctly predicted positive samples over total predicted positives.
- Recall (Sensitivity): Correctly predicted positives over total actual positives.
- F1-Score: Harmonic mean of precision and recall.
- AUC-ROC: Area under the Receiver Operating Characteristic curve.
- Balanced Accuracy: Average of recall obtained on each class.

To optimize these metrics, we selected the best threshold for each model using the validation set.

5.2 Results Summary

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Balanced Accuracy	Best Threshold
MobileNetV2	~0.65	0.88	0.61 (Sick) / 0.71 (Healthy)	0.72 (Sick) / 0.47 (Healthy)	~0.71	~0.66	0.48
ResNet50	~0.76	0.89	0.77 (Sick) / 0.68	0.83 (Sick) / 0.55 (Healthy)		~0.72	0.16
EfficientNetB0	~0.76	0.91	0.70 (Sick) / 0.78 (Healthy)	0.79 (Sick) / 0.55 (Healthy)	~0.80	~0.74	0.35

Note: All metrics correspond to the test set using the optimized threshold for each model.

5.3 Confusion Matrices

To visualize class-wise performance, we present the normalized confusion matrices for each model:

MobileNetV2

	Predicted Diseased	Predicted Healthy
Actual Diseased	0.61	0.39
Actual Healthy	0.29	0.71

ResNet50

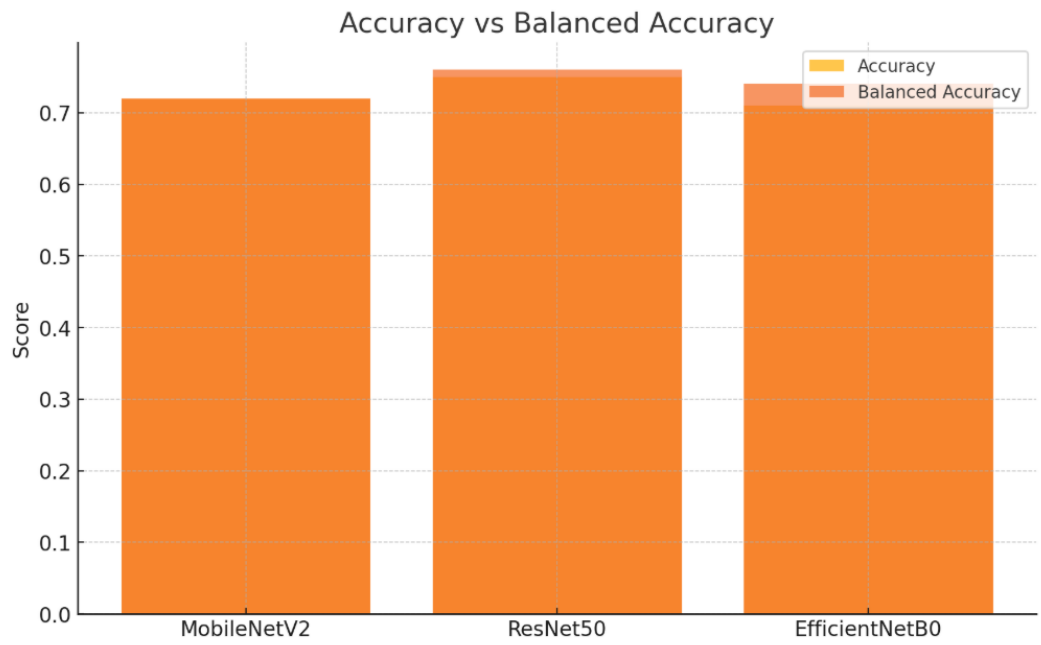
	Predicted Diseased	Predicted Healthy
Actual Diseased	0.77	0.23
Actual Healthy	0.32	0.68

EfficientNetB0

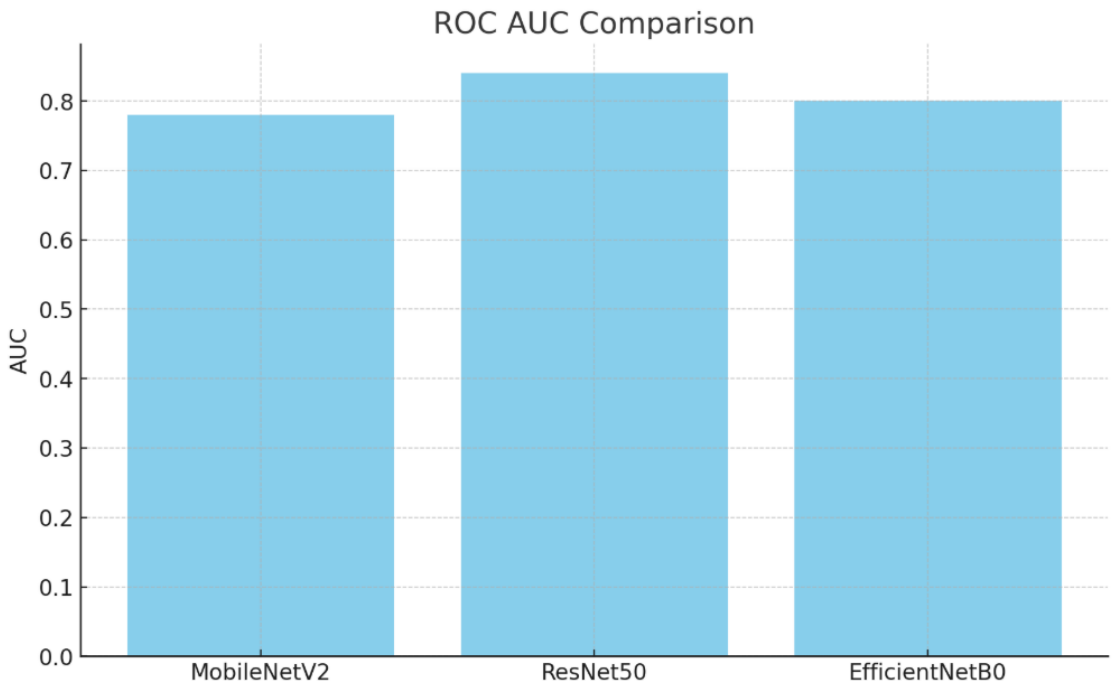
	Predicted Diseased	Predicted Healthy
Actual Diseased	0.70	0.30
Actual Healthy	0.22	0.78

5.4 Visual Comparisons

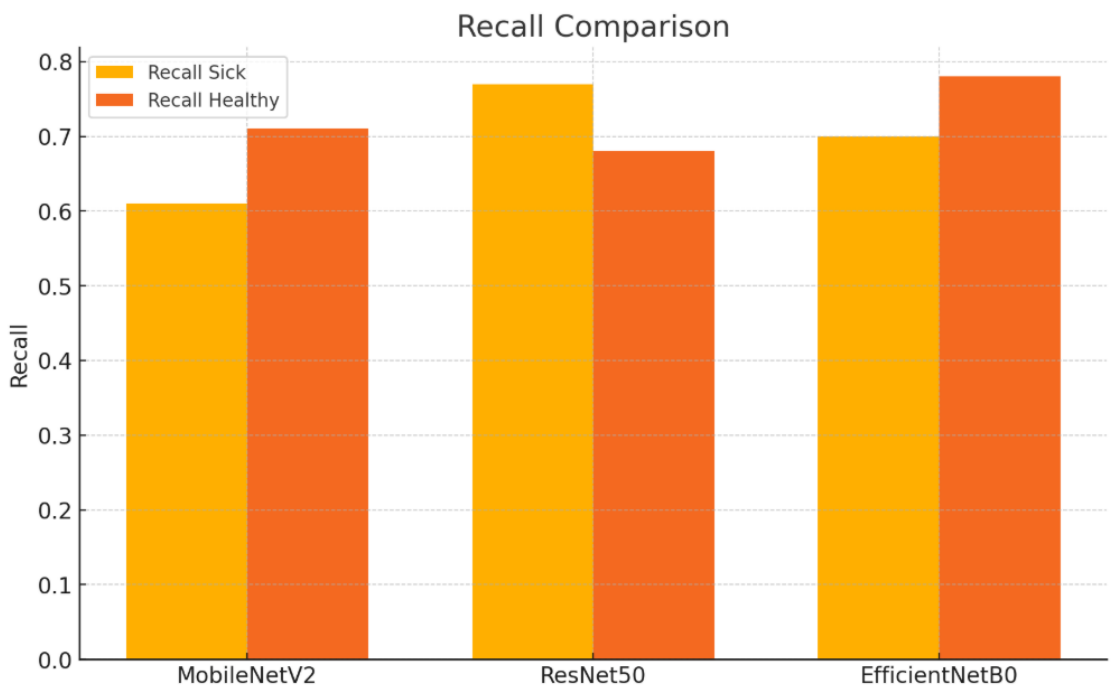
- Accuracy vs. Balanced Accuracy



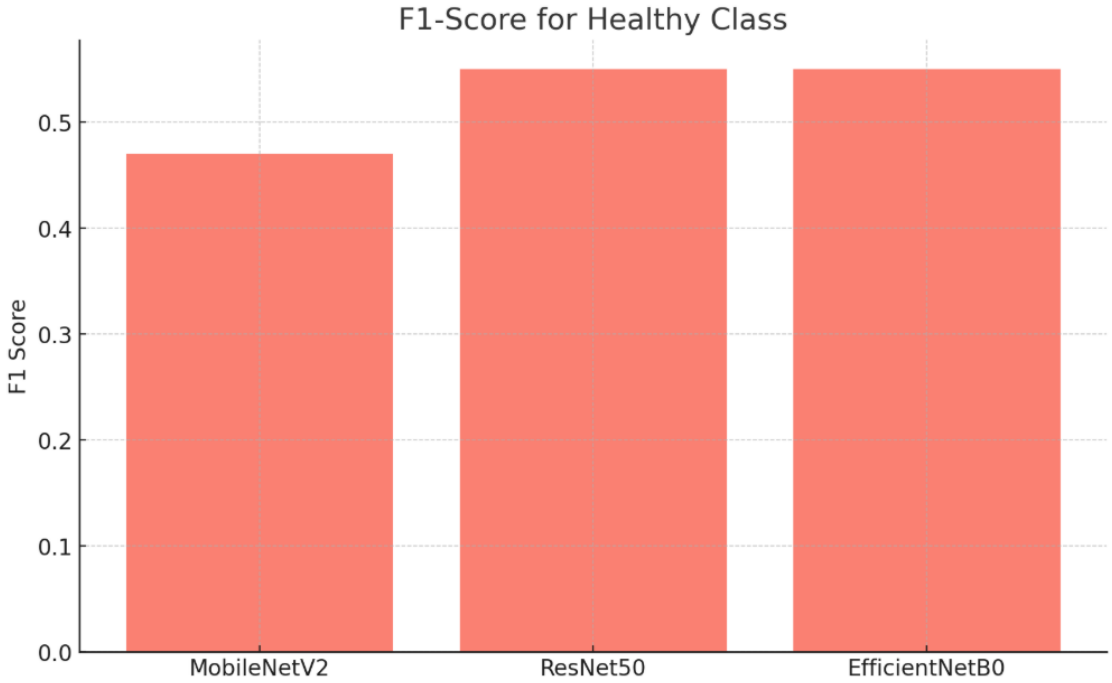
- AUC Comparison



- Recall for each class



- F1-Score for Healthy Class



5.5 Summary and Insights

EfficientNetB0 achieved the **best overall performance**, particularly in terms of balanced recall across both classes (Sick: 0.70, Healthy: 0.78) and F1-score stability, making it a strong candidate for clinical deployment.

ResNet50 proved to be a **robust and reliable model**, especially effective at identifying diseased cases (Recall: 0.77, F1: 0.83), though with slightly lower specificity.

MobileNetV2, while lightweight and computationally efficient, exhibited **lower generalization capacity**, particularly for the diseased class (Recall: 0.61), suggesting it's less reliable for high-stakes clinical scenarios without further tuning.

All models initially faced issues with overfitting and class imbalance, which were mitigated through:

- Targeted **fine-tuning** of pretrained layers
- **Data augmentation** to expand variability
- **Custom threshold optimization**, crucial to adjust the trade-off between sensitivity and specificity

The use of segmentation masks was key to improving performance, as it allowed the models to focus exclusively on kidney structures, filtering out irrelevant background noise.

Balanced Accuracy and **F1-scores** were preferred over raw accuracy due to the dataset imbalance, ensuring both classes were fairly evaluated.

6. Discussion and Conclusion

This project tackled the challenge of building a reliable kidney ultrasound image classifier, capable of distinguishing between healthy and diseased cases, using a relatively small and imbalanced dataset of **1,985 images** (1,535 diseased and 450 healthy). Through extensive experimentation with transfer learning, fine-tuning strategies, threshold optimization, and preprocessing techniques (including segmentation masks), we trained and evaluated three well-known convolutional neural network architectures: **ResNet50**, **MobileNetV2**, and **EfficientNetB0**.

After systematic fine-tuning and validation, the final performance metrics for each model are summarized below:

Comparative Results Table (Final Thresholds Applied)

Model	Threshold	Precision (Sick)	Recall (Sick)	F1 (Sick)	Precision (Healthy)	Recall (Healthy)	F1 (Healthy)	ROC AUC
ResNet50	0.16	0.89	0.77	0.83	0.46	0.68	0.55	
MobileNetV2	0.48	0.88	0.61	0.72	0.35	0.71	0.47	0.71
EfficientNetB0	0.13	0.91	0.70	0.79	0.43	0.78	0.55	0.80

Key Findings

EfficientNetB0 delivered the most balanced results overall. With the best F1-score for healthy patients (0.55), strong recall across both classes, and a ROC AUC of approximately 0.80, it proved to be the most effective model in terms of clinical sensitivity and generalization.

ResNet50 showed excellent performance in detecting diseased patients (F1 = 0.83, recall = 0.77), making it a very strong and reliable choice for minimizing false negatives—essential in medical diagnostics.

MobileNetV2, while faster and more lightweight, demonstrated more limited generalization. It tended to over-classify healthy cases as sick, which affected its overall F1 performance and AUC. Nevertheless, it still achieved respectable precision and recall on the diseased class.

6.2 Limitations

Despite encouraging results, the current system has some limitations:

- The dataset is limited to **2D grayscale ultrasound images**, without other patient metadata.
- There may be **variability in image quality and resolution** due to different equipment or acquisition conditions.
- The models still **struggle to perfectly generalize** to unseen healthy cases due to the data imbalance and the few images that compose the dataset.

These factors limit generalizability and underline the importance of external clinical validation.

6.3 Future work

We identify several directions for enhancing this project:

1. **Dataset Expansion**
    - Increase the number of healthy samples, or apply more advanced generative data augmentation (e.g., GANs).
2. **Model Explainability**
    - Integrate SHAP or LIME explanations for further transparency, or techniques as GradCam.
3. **Ensemble Methods**
    - Combine MobileNetV2, ResNet50, and EfficientNetB0 into an ensemble to leverage their complementary strengths.
4. **Clinical Validation**
    - Collaborate with healthcare institutions to test the system on real-world clinical data.
5. **Web App Integration**
    - Expand the deployed web application with image quality checks, user feedback collection, and multilingual support.
6. **Real-time Deployment**
    - Export models to ONNX or TensorFlow Lite for integration into ultrasound machines or portable diagnostic devices.